RETENTION INTELLIGENCE REVIEW

Churn & Retention Intelligence

Where the subscription book is losing customers and revenue, why those losses cluster where they do, and which accounts to defend first.

Analytical period Feb 2022 to Feb 2026 · snapshot reference date 1 March 2026
Book of 3,500 lifetime accounts · 2,517 active at snapshot · $1,117,779 active monthly recurring revenue

Retention Analytics · Companion deliverable to the self-contained executive dashboard (`executive-retention-command-center.html`).
Findings are decision-support only, not audited financial results.

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SECTION 01

Executive summary

The book carries 3,500 accounts. As of the 1 March 2026 snapshot, 983 have churned and 2,517 remain active, billing $1,117,779 of monthly recurring revenue. The headline customer churn rate is 28.1 percent; the monthly-value loss share is materially lower at 19.5 percent. That spread is the story. The book is losing too many logos, but the loss is skewed toward lower-value accounts. The response should therefore be selective, not universal.

28.1%	19.5%	227	$143k
CUMULATIVE CUSTOMER CHURN	CUMULATIVE REVENUE-LOSS SHARE	CRITICAL + HIGH-RISK ACCOUNTS	WEIGHTED MRR EXPOSURE

Management has two jobs. First, protect the open accounts where behavioural risk and revenue matter: 227 named critical and high-risk accounts holding $125,734 of MRR. Second, reduce the inflow of fragile demand. Affiliate and Paid Search generate 34.7 percent of all accounts but 45.3 percent of churned accounts. Customer Success can work the first problem. Revenue leadership owns the second.

> *Customer Success can work the first problem. Revenue leadership owns the second.*

Three findings drive the recommendation. First, churn is concentrated in the low-commitment end of the book. Startup accounts churn at 36.8 percent versus 16.6 percent for Enterprise. Affiliate-sourced accounts churn at 41.6 percent versus 18.3 percent for Partner. Basic-plan accounts churn at 38.1 percent versus 12.9 percent for Enterprise plans. These descriptive cuts make acquisition mix and entry-tier economics the first commercial hypotheses to test; they do not isolate causal effects.

Second, pre-churn account health separates leavers from stayers with commercially useful clarity. Accounts carrying a low net promoter score churn at 47 percent against 22 percent for accounts without that flag. A heavy recent support burden lifts churn to 40 percent. Failed payments, weak feature adoption, and declining usage each mark out populations that leave at several times the book rate. These signals surface before the account closes. That is what makes them useful for prevention, not just for post-mortem accounting.

Third, the revenue exposure is concentrated enough for named-account management. The top 5 percent of churned accounts by value account for 30 percent of lost monthly revenue; the top 20 percent account for 61 percent. On the forward book, the critical and high tiers hold $125,734 of MRR. Four intervention queues cover the material exposure: Renewal Save Desk, Payment Rescue, Adoption Reactivation, and Service Recovery. Sequence them by weighted exposure; payment rescue can run in parallel as a distinct billing workflow.

Executive decision	Evidence	Required move
Do not run a base-wide save campaign	227 high/critical accounts hold $125,734 of MRR.	Assign named owners to the high-risk queue and protect capacity from low-signal accounts.
Treat acquisition mix as a retention lever	Affiliate + Paid Search: 34.7% of accounts, 45.3% of churned accounts.	Test a marginal channel-mix shift and retain it only if live cohort retention improves without weakening acquisition economics.
Separate human and automated motions	765 active accounts carry multiple distress signals; 821 carry none.	Use human outreach for multi-signal value accounts; use scaled nudges for single-signal usage decline.
Govern by outcomes, not activity	Intervention queues cover $351,889 of MRR in scope.	Track saved MRR versus control, recovered payment MRR, and 6-month retention by source.

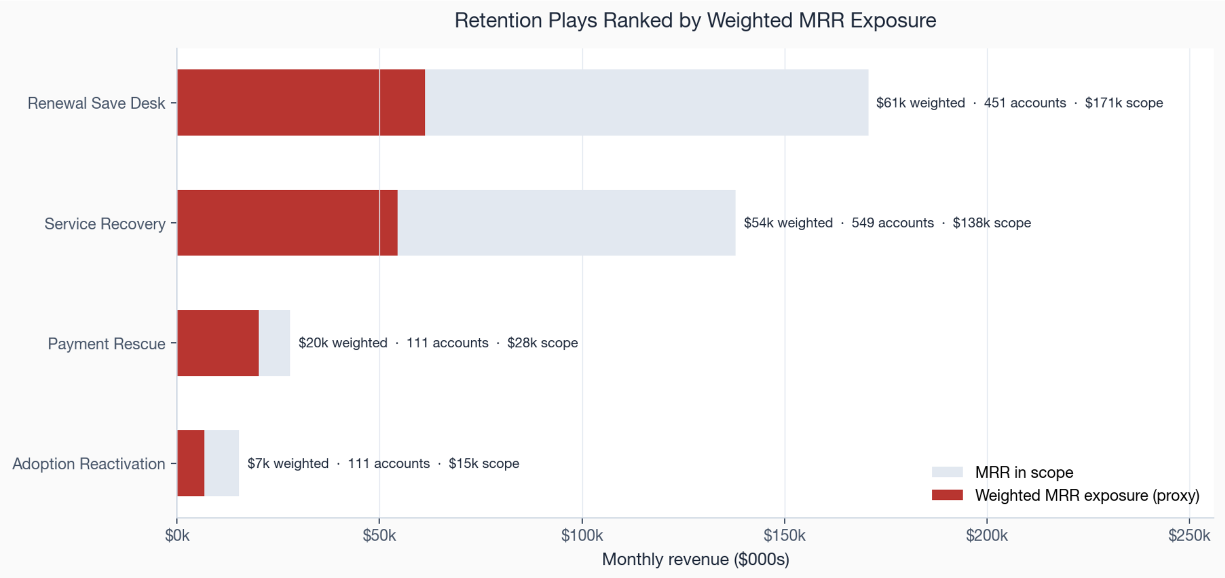

Figure 1. Retention plays ranked by weighted MRR exposure. The shaded bar is the monthly revenue in scope for each play; the solid bar is that revenue weighted by the historical churn rate of the accounts it targets.

The sequence follows weighted exposure and operational separability. Stand up the renewal save desk first: it carries the largest weighted exposure and the accounts are identifiable now. Fix the payment-failure path at the same time: the queue is small, the implementation is low-complexity, and affected accounts churn at more than three times the rate of accounts that pay cleanly. Next, test whether a marginal channel-mix shift away from Affiliate and Paid Search improves retention after controlling for segment, plan, and acquisition economics.

The weekly operating view should track new critical and high-risk accounts, payment recoveries, usage-trend recoveries, and queue aging. The monthly executive view should track two portfolio levers: lower weighted exposure in the open-account queue and a lower weak-channel share in new cohorts. Held-out outcomes, not activity volume, should determine whether either intervention scales.

One caveat governs everything that follows. This analysis runs on deterministic synthetic data. The transparent policy score prioritises operations, while a separate calibrated model estimates 90-day churn probability on held-out time periods. Both are internally consistent; their magnitudes remain illustrative until live outcomes replace the simulation. Section 10 states the limitations in full.

SECTION 02

Context and objectives

This review gives revenue leadership and customer success a common retention baseline: the scale and timing of loss, the populations where it concentrates, the observable signals that precede it, and a value-adjusted queue of open accounts. A reconciled MRR movement ledger, acquisition spend, direct service cost, calibrated probabilities, and randomized holdouts extend that baseline into unit economics and controlled measurement. The values remain analytical proxies rather than audited accounts or observed treatment effects.

The book under review

The dataset is the complete account history of a business-to-business software company: 3,500 accounts acquired across four customer segments (Enterprise, Mid-Market, SMB, and Startup), four regions (North America, Europe, APAC, and LATAM), six acquisition channels, and four plan tiers. Monthly activity runs from Feb 2022 to Feb 2026. At the snapshot date, 2,517 accounts are active and 983 have closed. The active book bills $1,117,779 per month. Each account carries the behavioural signals a real customer success team would have in front of it: recent product usage and its trend, feature adoption, support ticket volume, net promoter score, and payment history.

The decisions this report supports

This work answers five questions a retention owner has to make calls on. It does not produce a general description of the data:

- **How is the book trending?** Are monthly customer and revenue churn rising, flat, or falling, and is the loss weighted toward high-value or low-value accounts?
- **Where does churn concentrate?** Which segments, plans, channels, and regions carry a disproportionate share of the loss, and are recent cohorts retaining better or worse than older ones?
- **What separates leavers from stayers?** Which observable account-health signals are most strongly associated with churn, and by how much do they lift the rate?
- **Where is revenue exposed?** How much current MRR sits behind each pattern, and is the loss concentrated in a small set of high-value accounts or spread thinly across many small ones?
- **What should be done first?** Which intervention queues carry the largest weighted exposure, and in what order should they be resourced?

Everything in the findings maps back to one of these five questions. Where the evidence is associative rather than causal, the report says so plainly instead of dressing a correlation as a cause.

SECTION 03

Data and methodology

This section describes where the numbers come from and how each metric is defined, so that any figure in the report can be traced back to a rule rather than a judgement call. The pipeline is deterministic: it runs from a fixed seed, every artifact is reconstructible, and the analytical tables are governed by data contracts that are checked before any output is released.

Data sources

Four raw tables feed the analysis. A customer table holds the acquisition attributes (segment, region, channel, signup date). A subscription table holds the plan, the monthly recurring revenue, and the start and end dates that define whether and when an account churned. A weekly product-usage table records sessions and feature interactions. A payment table records billing events and failures. These are joined into one per-account feature row measured either at the account's churn date or, for active accounts, at the snapshot reference date, so that every account is observed at a comparable point in its own life rather than at an arbitrary calendar date.

Layer	Artifact	Grain	Role in this report
Raw	customers, subscriptions, usage, payments	event / account	Source of truth for all downstream tables
Features	`customer_retention_features.csv`	one row per account	Behavioural signals at churn or snapshot
Analysis	`main_analysis_*, churn_by_*`	aggregate	Trends, driver ranking, revenue at risk
Risk	`customer_risk_scores.csv`	active account	Tiered priority queue and recommended action
Cohort	`cohort_retention_table.csv`	cohort × month	Retention curves and heatmap

How the synthetic data is built

The data is generated, not collected, and it matters that the reader knows what that means for the findings. The generator assigns each account a latent churn propensity driven by its segment, plan, channel, and a set of behavioural processes, then simulates usage, support, payment, and survey events consistent with that propensity. The result is a book in which the relationships between account health and churn are real and stable but designed, not discovered. The report treats the magnitudes as illustrative. What transfers to a real book is the metric definitions, the analytical framework, and the prioritisation logic.

Metric definitions

Several numbers in this report look similar and are easily confused, so each is pinned to an explicit definition here and used consistently throughout.

Metric	Definition
Cumulative customer churn	Closed accounts as a share of all accounts ever acquired. A lifetime, stock measure. Reads 28 percent here.
Monthly customer churn rate	Accounts that close in a month divided by accounts active at the start of that month. A flow measure, averaging 1.4 percent over the window.
Cumulative churn share (by group)	Within a segment, plan, channel, or region, the share of that group's accounts that have closed.
MRR at risk	Current monthly recurring revenue billed by active accounts that carry the at-risk flag at the snapshot.
Churn rate lift	Churn rate inside a signal divided by churn rate outside it. A multiple, not a percentage.
Weighted MRR exposure	Revenue in an intervention queue multiplied by the historical churn rate of the accounts it targets. A heuristic for sizing, not a forecast.
Retention priority score	A transparent index combining behavioural risk and customer value, used only to rank the queue.

Two of these deserve emphasis. Cumulative customer churn and the monthly churn rate are different lenses on the same book and will not match: the first asks what share of everyone ever signed has left, the second asks how fast the active book is losing accounts right now. Both appear in this report and neither is wrong. Revenue loss throughout uses a monthly-value proxy, the account's recurring monthly revenue at the point it left, rather than a full contract or lifetime value, which the synthetic data does not model.

Governance and release gates

No figure in this report or the dashboard is released until the pipeline passes its governance gates. The artifacts are checked against data contracts that assert the shape, types, ranges, and referential integrity of every table, and against a final validation suite covering data quality, metric correctness, analytical integrity, the dashboard render, and release policy. At the current build, all 57 of 57 checks pass. That is what licenses decision-support claims instead of screening-grade ones. The gate breakdown follows. The full check log lives in `outputs/tables` and regenerates on every run.

Validation gate	Checks passing
Analytical Integrity	6 / 6
Dashboard Review	14 / 14
Data Quality	9 / 9
Governance & Release	7 / 7
Metric Correctness	10 / 10
Strategic Expansions	11 / 11

The release policy is deliberately conservative about what these passes entitle the analysis to claim. Clearing every gate makes the output technically valid and analytically acceptable for decision support with explicit caveats. It does not make it committee-grade, because the synthetic-data limitation in Section 10 is an unresolved caveat by construction. The governance layer encodes that distinction rather than leaving it to the reader's goodwill.

Evidence standard applied in the body

The body of the report uses three levels of evidence. Descriptive cuts are decision-useful when the same pattern repeats across segment, plan, channel, and value. Behavioural signals are decision-useful when they separate leavers from stayers and also identify active accounts that can still be worked. Recommendations are decision-useful only when they combine both: a population the business can identify, a revenue stake large enough to matter, and a metric that can validate whether the intervention changed outcomes rather than activity.

Evidence question	Report test	What would weaken the read
Is the pattern real?	The finding repeats across rate, revenue, and mix cuts rather than appearing in one chart only.	A single small segment, a right-censored period, or a mix effect that disappears when read at fixed cohort age.
Is the pattern actionable?	The affected accounts can be named before churn and routed to a clear commercial owner.	A signal that is visible only after churn, or a population too broad for the team to treat differently.
Is the value material?	The queue carries current MRR or lost monthly value large enough to change prioritisation.	High churn rate with negligible account count or revenue scope.
Can the action be validated?	The recommendation names a success measure and admits where a control or holdout is needed.	ROI asserted from historical association without an experiment or before/after design.

SECTION 04

Analytical framework

The analysis moves through six lenses, each narrowing the question. The trend shows whether the book is getting better or worse in aggregate. Decomposing the loss by commercial attribute and by acquisition cohort shows where it lives. Account-level behaviour reveals what the leaving accounts had in common before they left. Attaching revenue and a priority score to each pattern turns the diagnosis into a ranked operating queue. A reconciled MRR bridge adds retention-adjusted economics; a calibrated model, randomized holdout, and transition monitor add prediction, measurement, and operational control.

Associative, not causal

Every relationship in this report is associative. When the report says that accounts with a low net promoter score churn at a far higher rate, it means the two co-occur in the data, not that a low score causes the departure. The low score and the departure may both be downstream of a third thing, a poor onboarding or a product gap, and the data here cannot separate those. Establishing cause requires controlled experiments, holding out a population from an intervention and measuring the difference. The reference pipeline implements that assignment and estimator, but its forward outcomes remain simulated. The driver ranking tells the operator what to test, weighted exposure sets priority, and live holdout outcomes must settle efficacy.

The decomposition step is not cosmetic. A single book-wide churn rate can move in the opposite direction to every one of its parts if the mix of those parts is shifting, the trap known as Simpson's paradox. A business whose every segment is improving can post a worsening aggregate simply by acquiring more of its weakest segment. Cutting the rate by segment, plan, channel, and region guards against reading a mix shift as a performance change, and it is why the cohort and concentration findings are presented next to the aggregate trend rather than in place of it.

The risk index

Active accounts are scored on two axes and ranked on their product. The first axis is behavioural risk, built from the same pre-churn signals examined in Section 08: usage trend, feature adoption, support burden, net promoter score, and payment failures. The second axis is customer value, built from current monthly revenue and tenure. An account is worth working when it scores high on both, a high-value account showing real distress, rather than on either alone. The mean risk score across the 2,517 scored accounts is 13.3 on a 0-to-100 scale, which is deliberately low: most of the book carries no material distress signal, and the index exists to surface the minority that does. Accounts fall into four tiers, critical, high, medium, and low, and each carries a single recommended action so the output is a worklist rather than a chart.

The index is transparent by design. Every input is a named, inspectable signal and the combination rule is a documented formula, not a black-box model. An operator can inspect why an account surfaced and challenge the underlying signal. Predictive accuracy has not been estimated here, so the index cannot be compared with a trained classifier until both are tested on time-separated outcomes.

SECTION 05

Findings: retention health and trend

Across the analytical window the book lost an average of 1.4 percent of its active accounts per month and 0.9 percent of its active revenue per month. Revenue churn running below customer churn is the important signal: the average departing account is worth less than the average retained account. The book is losing the low end faster than the revenue spine. That is a better loss shape than the reverse, but it still creates avoidable acquisition demand. The later sections locate that low-end pressure.

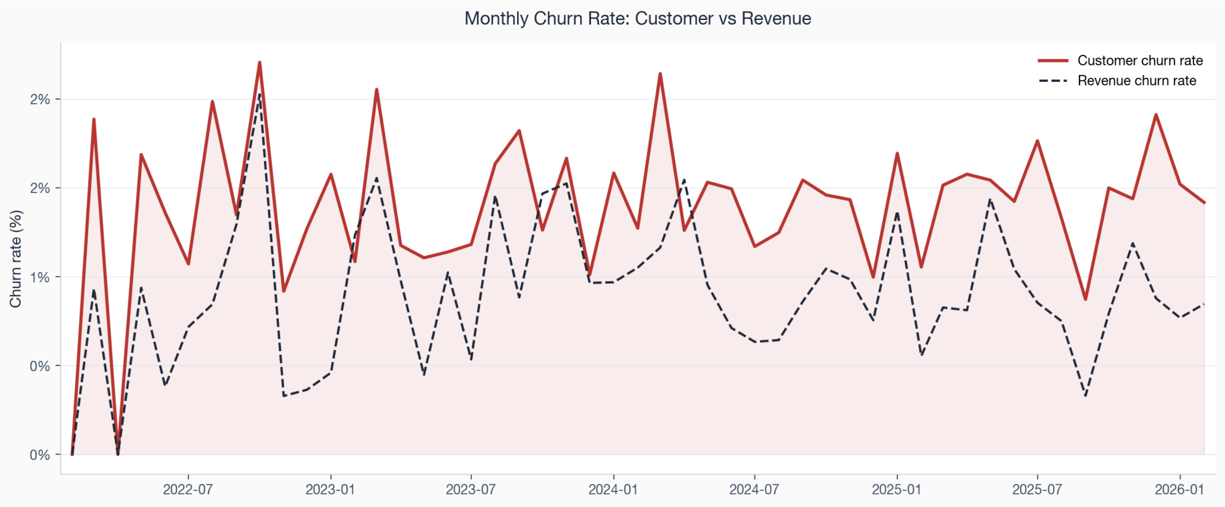

Figure 2. Monthly customer and revenue churn rates across the analytical window. The revenue line sits consistently below the customer line, the signature of low-value accounts leading the losses.

The average level is manageable; the recent direction needs attention. Monthly customer churn in the final observed month reads 1.4 percent against a prior nine-month average of 1.4 percent. The last three months sit above the trend that preceded them. Two readings are possible, and both matter. Operationally, the most recent quarter may be losing accounts faster than the settled baseline. From a measurement perspective, accounts closing near the snapshot date are mechanically over-represented because the window is right-censored, so the final months are not fully comparable to settled history. Section 10 returns to this. The right posture is weekly monitoring, not a broad emergency response.

The quantified acceleration is material on both count and value. The final three observed months average 1.6 percent customer churn, 0.2 percentage points above the prior nine months. Revenue churn averages 0.8 percent, -0.1 points above its prior nine-month baseline. Because the revenue increase is smaller than the customer-count increase, the recent spike is still dominated by lower-value accounts. That keeps the response focused: monitor the spike weekly, but do not let it pull scarce save-desk capacity away from the high-value account tail.

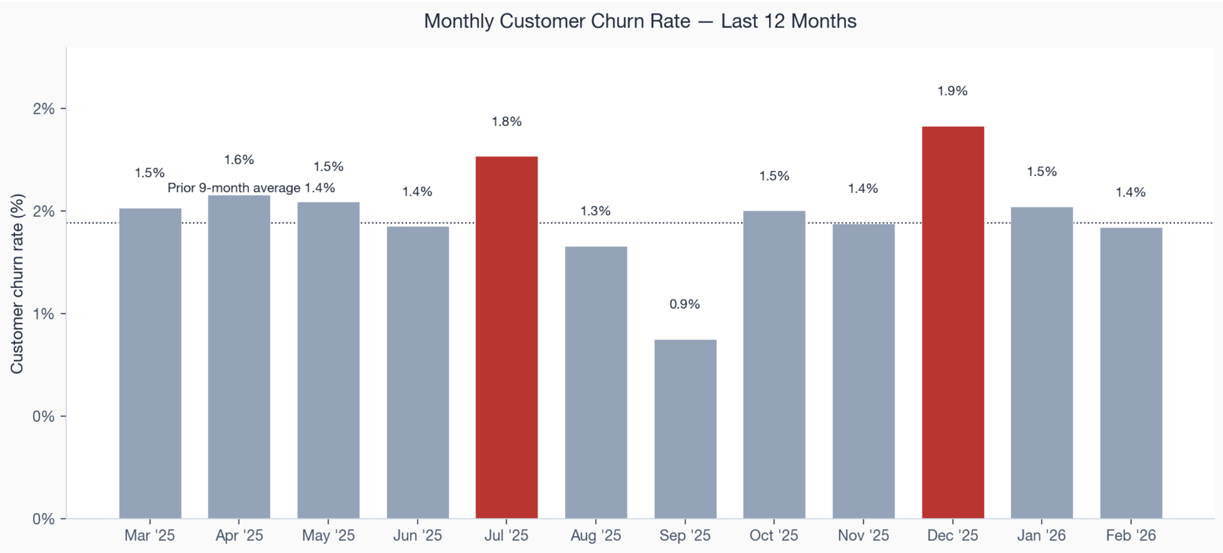

Figure 3. Monthly customer churn rate over the last twelve months. The final three months (highlighted) sit well above the prior nine-month average shown by the dotted line.

Underneath the rates, the active book has grown steadily in revenue terms to $1,117,779 per month at the snapshot, so the business is adding more than it is losing. That makes this a question of efficiency, not survival: every point of avoidable churn is a point the acquisition engine has to refill before the book grows, and the next sections show that a large share of the current loss is concentrated in places where management has practical levers.

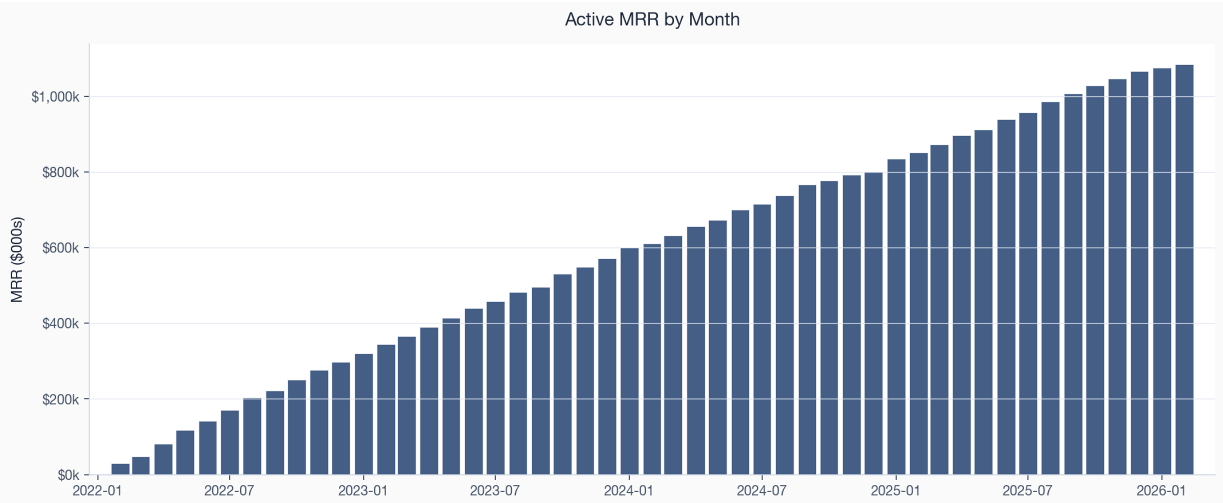

Figure 4. Active monthly recurring revenue by month. The book has compounded steadily; the retention task is to protect the base, not to rescue a business in decline.

The shape of the loss

The gap between customer churn (1.4 percent a month on average) and revenue churn (0.9 percent) is worth reading deliberately, because its sign decides the whole posture of a retention programme. When revenue churn runs below customer churn, the average departing account is worth less than the average retained one, and the book is upgrading its mix even as it loses count. That is the case here. The dangerous pattern is the reverse: a few large accounts leave, customer churn stays low, and revenue churn spikes. That pattern would demand immediate top-account defence. This book calls for a more segmented operating model: scaled treatment for the low-value tail, named-account coverage for high-value exceptions.

The gap does not establish unit economics. This dataset has no acquisition cost, gross margin, expansion, contraction, or remaining contract value, so it cannot estimate lifetime value or the economic benefit of retaining a specific account. It does show why logo and revenue retention must be read together: a stable revenue view can coexist with a material customer-count loss. The cohort and concentration findings that follow locate where that difference sits in the book.

SECTION 06

Findings: cohort retention

Aggregate churn rates hide whether the business is getting better at keeping the customers it acquires. Cohort analysis answers that by following each monthly intake of new accounts across its own life and comparing intakes at the same age. A fixed-age gap identifies a change in cohort quality; it does not distinguish acquisition mix, onboarding, product fit, or random variation without additional evidence.

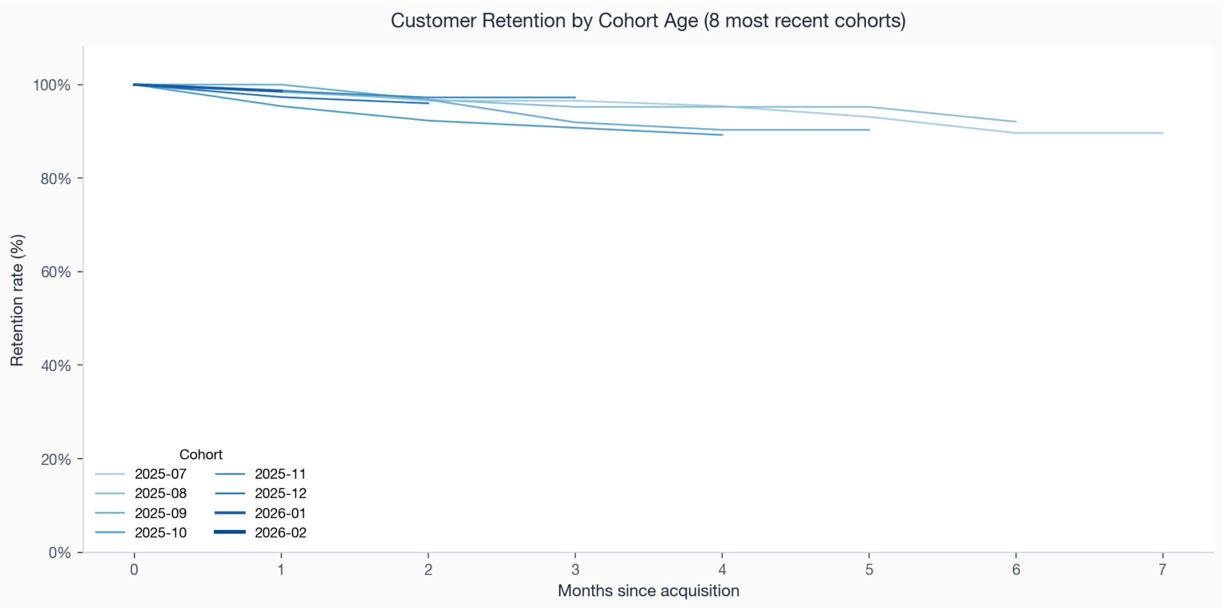

Figure 5. Retention by cohort age for the eight most recent monthly cohorts. Each line follows one acquisition month as its accounts age; the most recent cohorts are drawn most prominently.

Read at a fixed age, the curves bend the wrong way. Early cohorts hold almost all of their accounts through the first half-year. More recent cohorts start to separate downward at the same ages. The deterioration is modest in absolute terms, six-month retention still reads near 98 to 99 percent. The most recent cohorts are right-censored: they have not yet lived long enough to be compared fairly at the longer horizons, so the long-run gap should be read as a leading indicator to monitor rather than a settled fact.

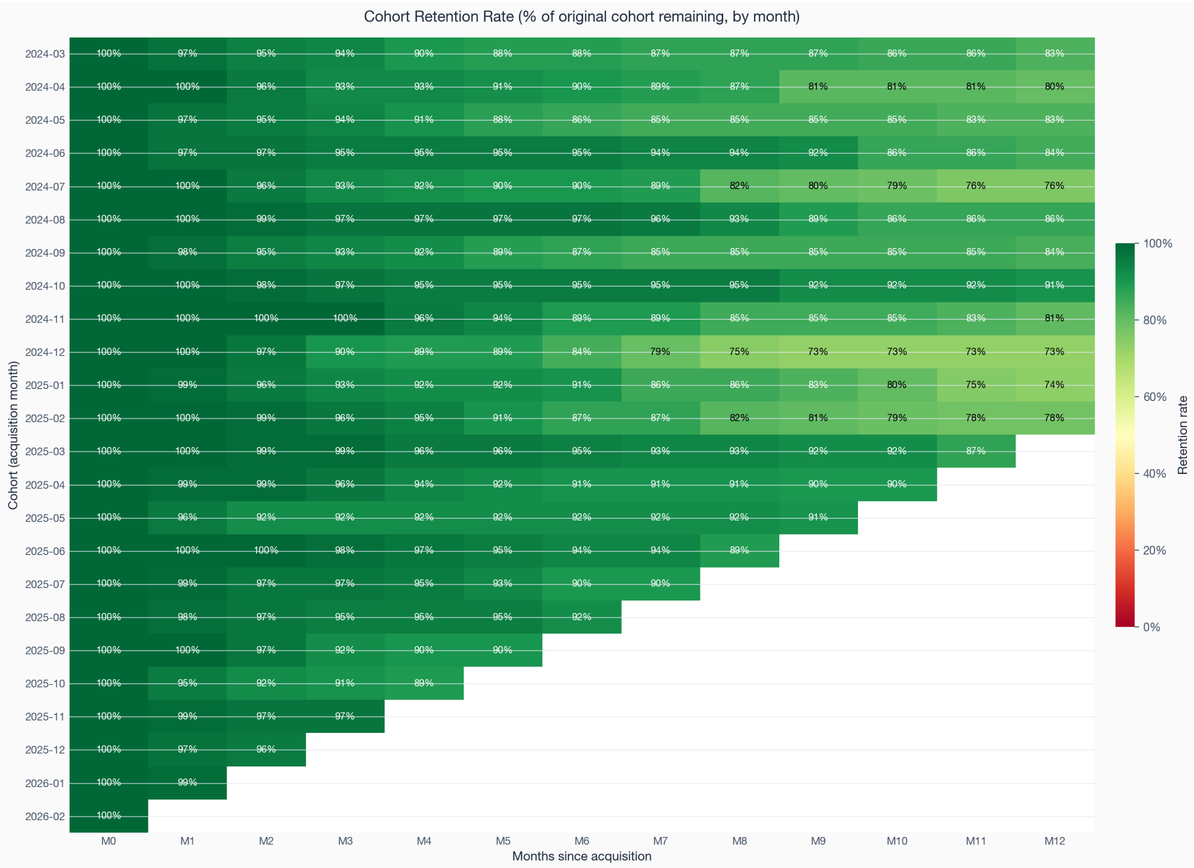

Figure 6. Cohort retention heatmap. Each row is an acquisition month, each column a month of age, and each cell the share of the original cohort still active. Green is higher retention, red is loss; the gradient down the early columns is where onboarding quality shows up.

The heatmap makes the pattern legible at a glance. The strongest retention sits in the upper rows, the older cohorts, and the early-age columns. As the eye moves down toward the recent cohorts, the early-age cells lose a little of their green. This is the same story the curves tell, shown as a surface. The chart supports monitoring a possible cohort-quality shift; it does not diagnose acquisition mix or onboarding as the cause. That question connects to the channel and segment findings in the next section.

The fixed-age read is the stricter test. It compares cohorts at the same age, so it does not penalise recent cohorts for not yet having long lives. On that basis the signal is not a single noisy month: recent cohorts trail early cohorts at month three, six, nine, and twelve. The logo gap is larger than the revenue gap at every age, confirming that the deterioration is concentrated in smaller accounts. The table below is why this pattern counts as a real operating signal, not yet a settled long-run outcome.

Age	Early logo ret.	Recent logo ret.	Logo delta	Early rev. ret.	Recent rev. ret.	Rev. delta
3m	96.2%	95.0%	-1.1 pp	98.1%	98.3%	0.2 pp
6m	92.5%	92.2%	-0.4 pp	95.6%	95.3%	-0.3 pp
9m	89.1%	84.8%	-4.3 pp	94.6%	93.0%	-1.6 pp
12m	86.8%	80.0%	-6.9 pp	93.4%	89.4%	-3.9 pp

One distinction matters for how this finding is used. Logo retention, the share of accounts still active, deteriorates a little faster than revenue retention, the share of original cohort revenue still billing, because the accounts leaving the recent cohorts are disproportionately the smaller ones. Six-month revenue retention holds near 99 percent even where logo retention slips below it. That is the same favourable shape seen in the aggregate trend. The cohort drift is a problem of small-account volume, not an erosion of the cohorts' revenue spine. The fix belongs upstream in acquisition and onboarding, not in a high-touch save motion aimed at the recent cohorts.

SECTION 07

Findings: where churn concentrates

This section turns the aggregate churn rate into an operating target. The book-wide rate averages together populations with very different economics, purchase intent, and support needs. Segment, plan, channel, and region cuts show where the leaving happens and whether the same weak populations appear from multiple angles.

By customer segment

Segment is the sharpest single cut. Startup accounts have lost 36.8 percent of their number and SMB accounts 32.2 percent, against 23.3 percent for Mid-Market and just 16.6 percent for Enterprise. The gradient is monotonic in account size: the smaller the customer, the more likely it is to leave. The same gradient runs through average revenue, where Enterprise accounts bill $910 a month against $170 for Startup, and through the health signals examined below. Small accounts churn more and are worth less individually. That is why the revenue line sits below the customer line in the trend.

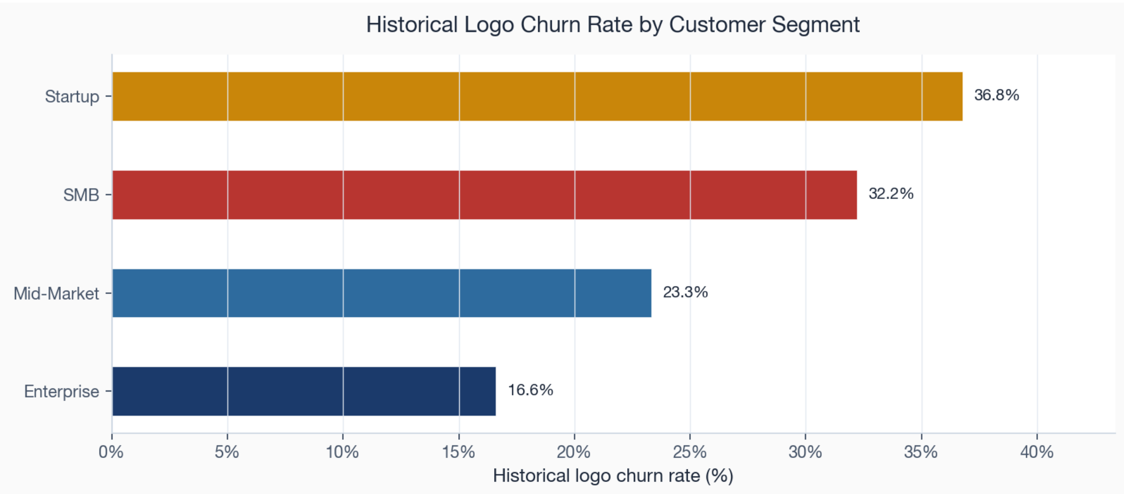

Figure 7. Cumulative churn share by customer segment. The gradient is monotonic in account size, from Enterprise at the low end to Startup at the high end.

The segment story goes beyond churn rates. It is about the whole health profile. The comparison below puts cumulative churn next to average net promoter score and the average usage trend for each segment. Enterprise accounts churn least, score highest on net promoter, and show usage that is still growing. Startup accounts invert all three. The signals move together rather than firing independently. That is what makes a single segment-level health read meaningful.

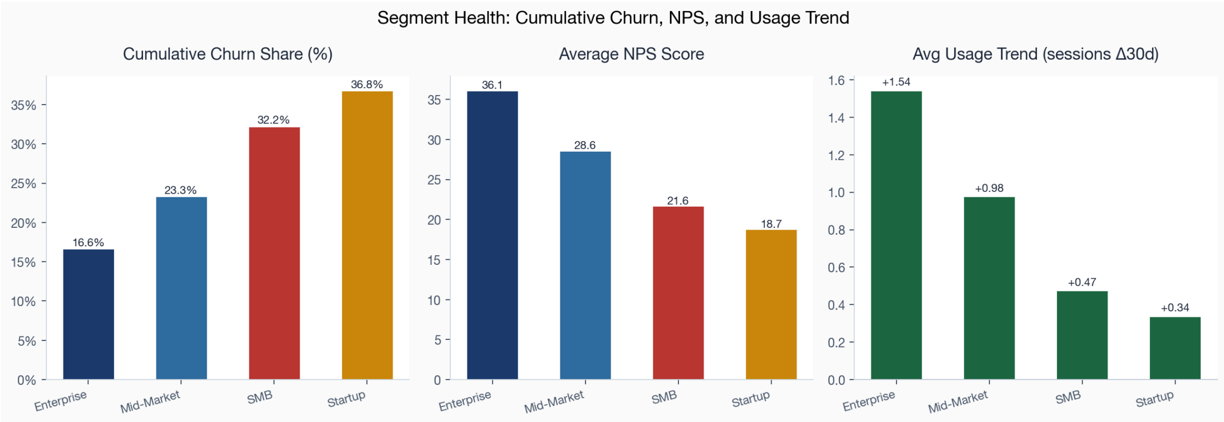

Figure 8. Segment health on three axes: cumulative churn share, average net promoter score, and average usage trend. A segment that is losing accounts is also the segment with weak sentiment and declining usage.

By plan type

Plan tier tells the same story in pricing terms. Basic-plan accounts churn at 38.1 percent, nearly ten times the 12.9 percent rate of Enterprise-plan accounts, with Growth and Pro stepping down in between. Lower-priced plans behave like lower-commitment purchases in this book. The commercial implication is not to abandon the entry tier: it feeds the funnel. It is to treat a Basic-plan signup as a different operating asset from a Pro-plan signup, with a faster activation path and earlier risk monitoring.

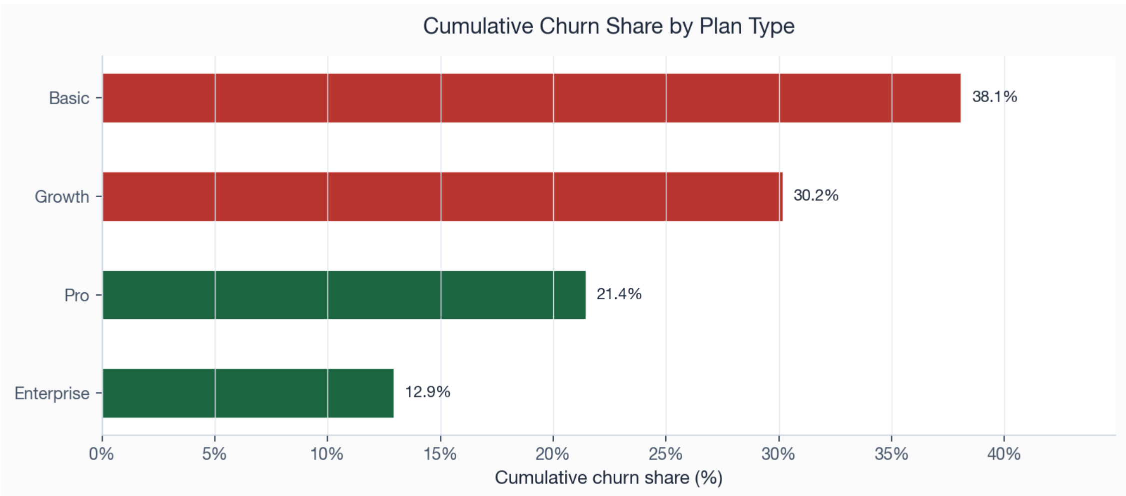

Figure 9. Cumulative churn share by plan type. Churn falls steeply as plan tier rises.

By acquisition channel

Channel is the clearest link between the retention read and spend decisions. Affiliate-sourced accounts churn at 41.6 percent and Paid Search at 34.3 percent, against 18.3 percent for Partner and 28.0 percent for Referral. Paid and affiliate demand is less durable than partner and referral demand in this run. The next step is a live cohort test that controls for segment, plan, CAC, and volume before changing channel budgets.

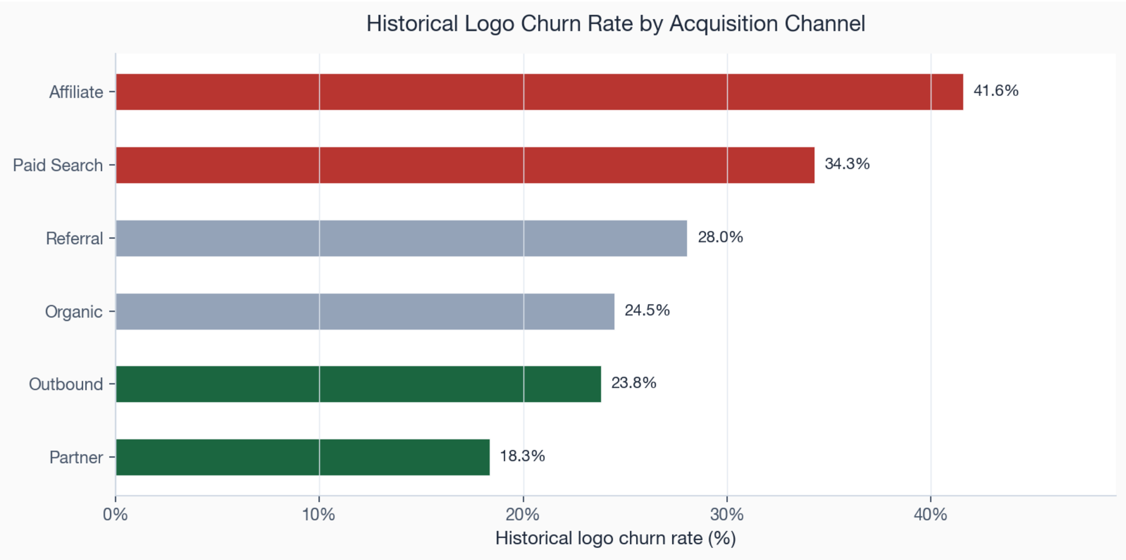

Figure 10. Cumulative churn share by acquisition channel. Paid and affiliate channels (red) deliver the least durable accounts; partner and referral channels (green) the most durable.

By region

Region is the weakest of the four cuts but not a flat one. LATAM accounts churn at 31.5 percent and APAC at 28.9 percent, both above the 28.1 percent book average, while North America sits below it at 27.0 percent. The spread is narrower than for segment or channel, and a good part of it probably reflects which segments and channels dominate each region rather than geography itself. Region is therefore a place to watch and to read alongside the other cuts, not a primary lever on its own.

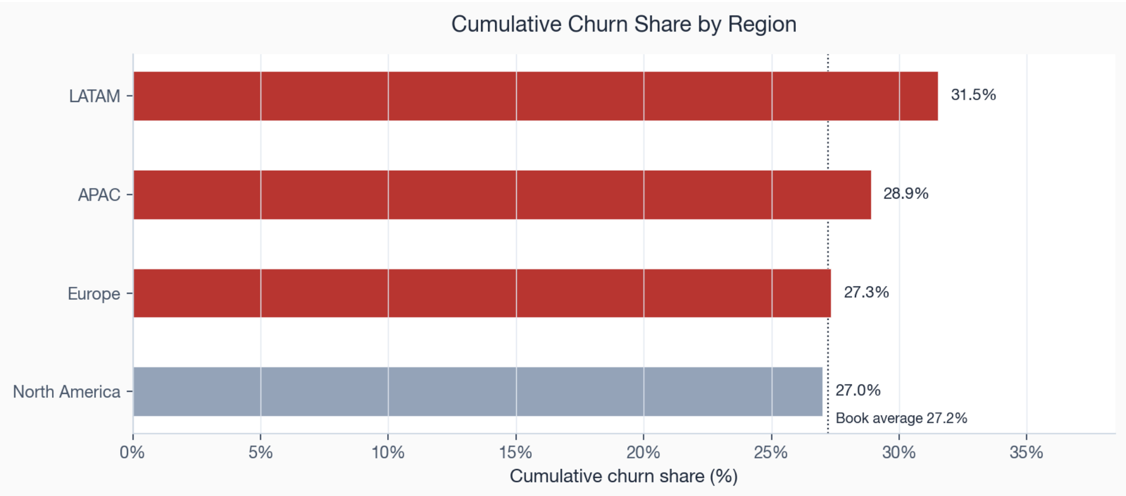

Figure 11. Cumulative churn share by region against the book average. LATAM and APAC run hot; North America runs cool.

Taken together, the four cuts describe an overlapping population rather than four unrelated problems. The accounts that leave are smaller, sit on cheaper plans, arrive more often through paid and affiliate channels, and are somewhat more concentrated outside North America. That overlap explains why revenue churn sits below customer churn and why recent cohorts have started to drift. The next section asks what those accounts looked like in the weeks before they left.

One word of caution carries over from the framework in Section 04. Because the four cuts overlap, they cannot simply be added. Affiliate traffic skews toward Startup and SMB accounts on Basic plans, so part of the channel effect is really the segment and plan effect wearing a channel label, and part of the regional spread is which segments and channels dominate each region. Separating the independent contribution of each attribute would take a model that holds the others constant. Budget reallocation should wait for that multivariate read and a controlled channel test.

The mix test below shows why this is still an executive issue rather than a statistical footnote. The weak commercial populations contribute more churn than their account share would imply, while the durable populations do the opposite. This is a screening signal for an upstream mix problem, not proof that channel or segment independently causes the difference.

Population	Account share	Churn share	Read
Startup + SMB	56.9%	67.9%	Low-end segments over-index in the churn pool.
Mid-Market + Enterprise	43.1%	32.1%	Durable segments under-index despite higher value.
Basic + Growth plans	61.8%	74.1%	Entry and mid tiers carry most logo churn.
Pro + Enterprise plans	38.2%	25.9%	Premium tiers are the revenue spine to defend.
Affiliate + Paid Search	34.7%	45.3%	Paid-intent sources bring disproportionate churn.
Partner + Referral	25.0%	20.0%	Trust-led sources bring more durable accounts.

SECTION 08

Findings: behavioural churn drivers

Commercial attributes say which accounts are structurally fragile. Behavioural signals say which accounts show current distress. The separation is wider than a live operating dataset would usually produce, which is a synthetic-data caveat, but the signals themselves are practical: recent net promoter score, support tickets, failed payments, feature adoption, and usage trend.

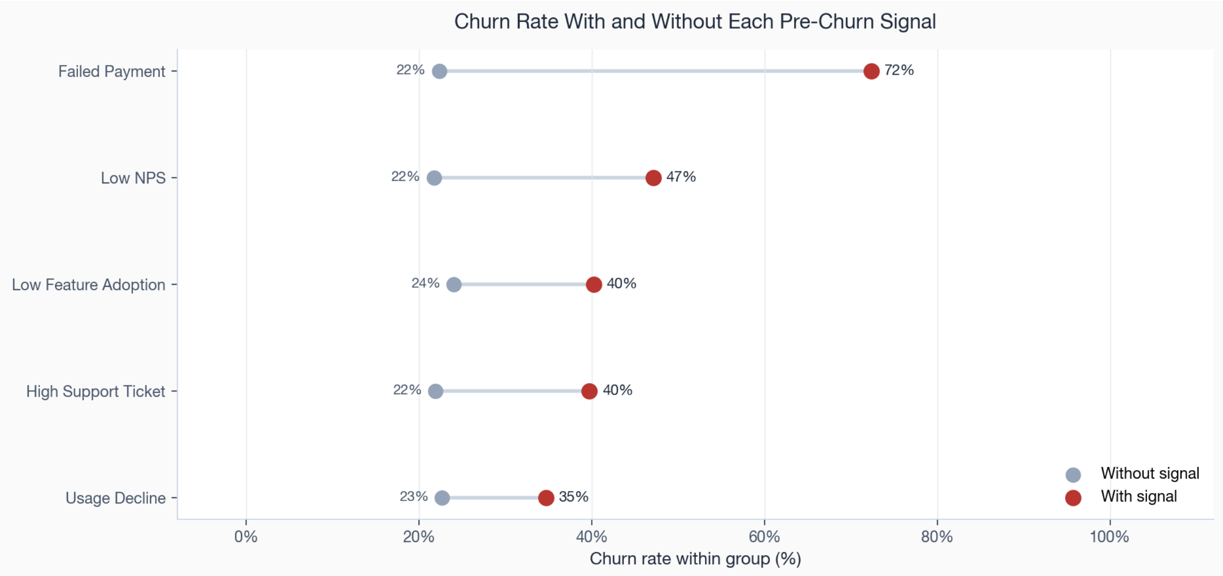

Figure 12. Churn rate for accounts with and without each pre-churn signal. The gap between the two dots is the decision signal: these flags split the book into populations with very different observed churn rates.

The separation is stark. Accounts carrying a low net promoter score churn at 47 percent against 22 percent for accounts without it. Accounts with a heavy support burden churn at 40 percent against 22 percent. Failed payments mark a population that churns at 72 percent, weak feature adoption 40 percent, and declining usage 35 percent. Expressed as a multiple, a low net promoter score lifts the churn rate by more than two times and a heavy support burden by more than one times relative to accounts without the flag.

These multiples are large partly because the comparison group, accounts without any distress flag, almost never churns in this book. That is a property of synthetic data and the report does not lean on the exact magnitudes. What survives the caveat is the ranking and the practical consequence: sentiment and support signals identify a small, high-risk population, while usage decline identifies a larger, lower-certainty population. The operating response should match that difference: human save motions for high-value accounts with severe signals, scaled nudges for the broad usage-decline tail.

Expressed as a multiple against accounts without each flag, the same five signals rank in a stable order. Low sentiment and high support burden sit far out at the top, payment failure in the middle, and weak adoption and usage decline closer to the no-lift line, though still well above it. The lift view and the absolute-separation view agree on the ranking and disagree only on emphasis. That makes the ordering more credible: it is not an artifact of one charting choice.

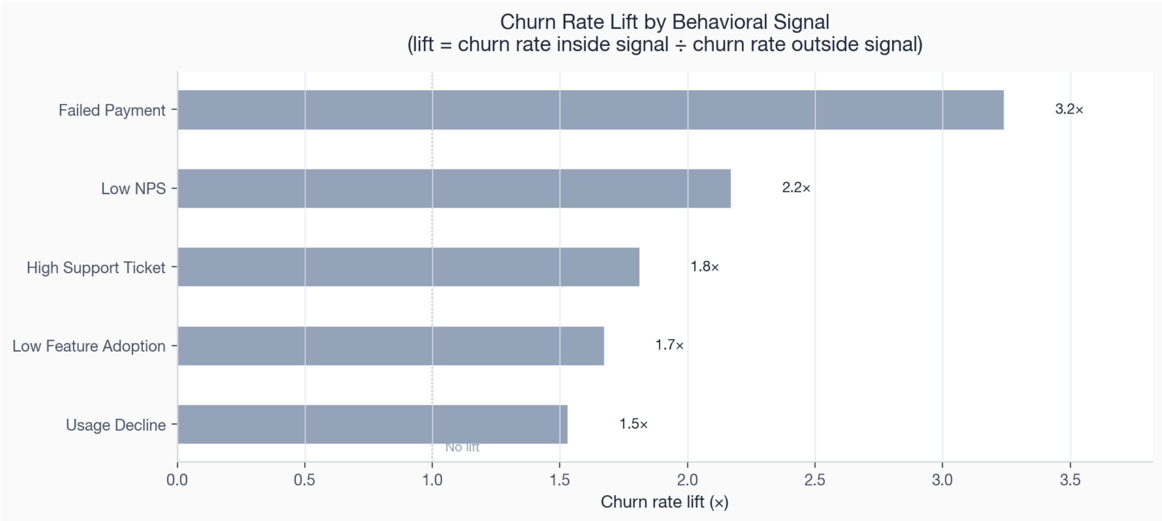

Figure 13. Churn rate lift by behavioural signal, the ratio of the churn rate inside each signal to the rate outside it. Bars past the dotted no-lift line mark signals that raise the rate; the strongest sit several times above it.

Ranking the drivers by money, not just by rate

A high lift on a tiny population is less important than a moderate lift on a large and valuable one. The ranking below therefore scores each driver by its excess MRR association: the monthly revenue tied to churned accounts carrying the signal, above what the book-average churn rate would predict. This reorders the list. Support burden and low sentiment stay near the top because they combine a high rate with real revenue. Usage decline rises sharply for a different reason: its lift is the smallest of the five behavioural signals, but it touches the most accounts and the most money.

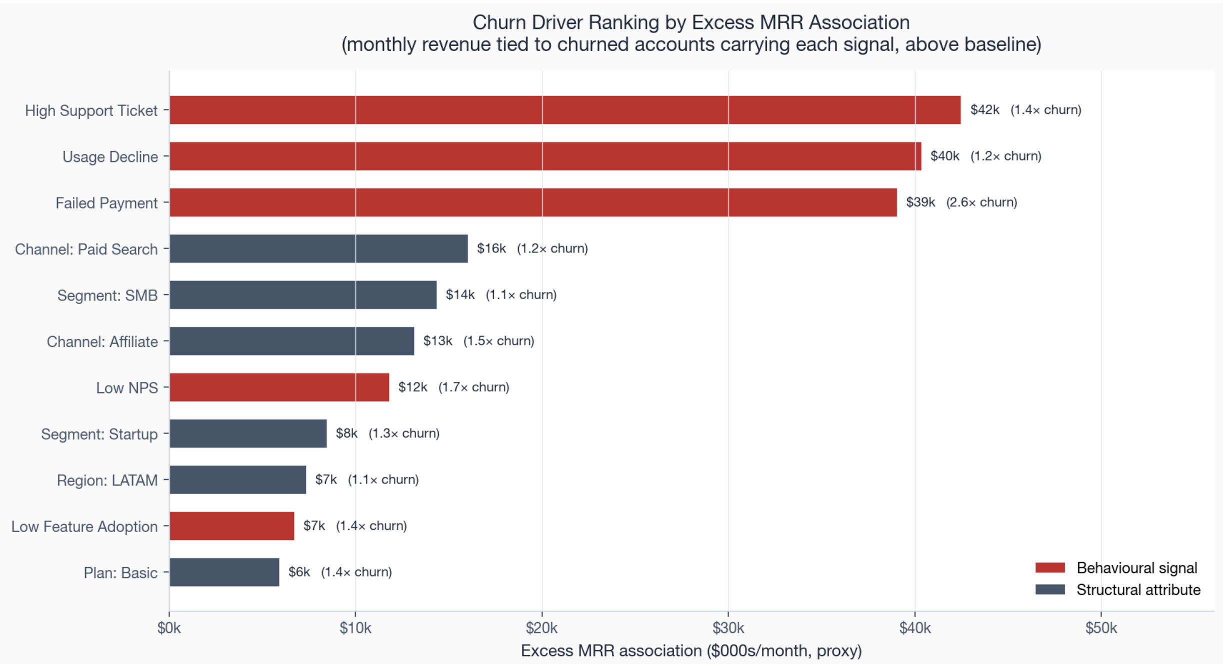

Figure 14. Churn drivers ranked by excess MRR association. Behavioural signals (red) dominate the structural attributes (slate); the label shows each driver's churn-rate multiple.

Driver	Accounts	Churn rate	Lift	Excess MRR assoc.
High Support Ticket	1,211	40%	1.4×	$42,462
Usage Decline	1,580	35%	1.2×	$40,345
Failed Payment	402	72%	2.6×	$39,024
Acquisition Channel=Paid Search	823	34%	1.2×	$16,021
Segment=Smb	1,416	32%	1.1×	$14,352

The five behavioural signals at the top of this table are the foundation of the risk index in Section 04 and of the intervention queues in Section 11. They define the eligible populations for the controlled intervention layer: each one maps to a plausible treatment and a revenue stake large enough to justify preserving a holdout.

The signals also compound. An account paying late and filing support tickets is in a different state from one carrying a single flag, and the risk index in Section 04 captures that by scoring on the full signal set. The save desk should work one ranked list, not five overlapping lists by signal. Multi-signal value accounts rise to the top; single-flag usage decline stays in the scaled-motion layer.

On the open book, 765 accounts carry two or more distress signals, representing $148,215 of current MRR. By contrast, 821 active accounts carry none of the five distress signals and account for $503,317 of current MRR. That split is important operationally. A broad outreach campaign would spend most of its effort on accounts with no current signal. A tiered queue reserves human coverage for the multi-signal tail and leaves low-signal accounts alone. The score's value is not that it finds a new signal; it prevents over-treatment.

SECTION 09

Findings: revenue at risk and concentration

Counting accounts tells you how many customers you are losing. It does not tell you how much money is leaving, or whether the loss is worth a targeted defence. This section attaches revenue to the risk and asks the question that informs the operating model: is the exposure concentrated enough to defend account by account, or so diffuse that only a broad programme could touch it?

The answer is that it is highly concentrated. Ranking churned accounts by value, the top tenth carries 44 percent of all lost monthly revenue, the top fifth carries 61 percent, and the top third carries 73 percent. By the halfway point of churned accounts by value, 86 percent of all lost monthly revenue is already accounted for. A small set of departures carries most of the lost monthly value. That concentration is the operating design point: a focused save desk working a few hundred named accounts can address the majority of the monthly-value exposure.

> *A small set of departures carries most of the lost monthly value.*

In dollars, the historical churn pool represents $270,784 of lost monthly value. The first 5 percent of churned accounts by value carry 30 percent of that loss, before the long tail even enters the discussion. This is why the response should be value-weighted. A logo-count target would push the team toward many small saves; a value-weighted target pushes it toward the accounts where one successful intervention changes the monthly run-rate.

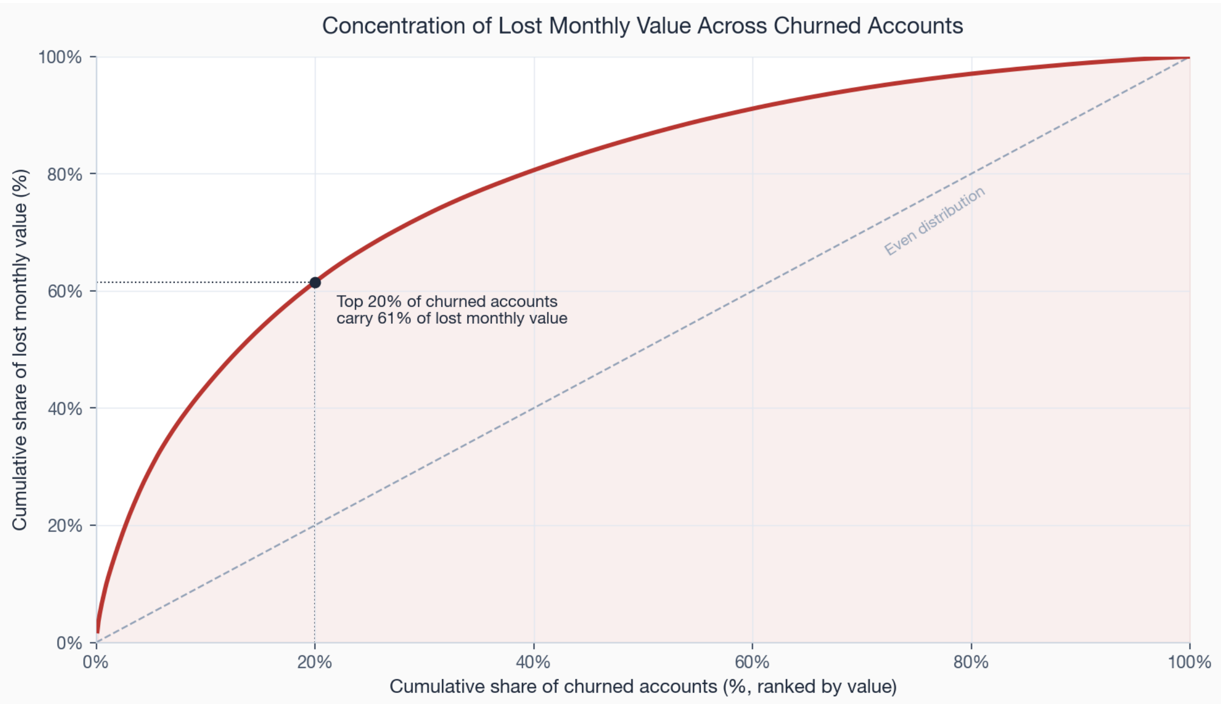

Figure 15. Concentration of lost monthly value across churned accounts. The curve bows far above the line of even distribution; the marked point shows the top fifth of accounts carrying the majority of the loss.

Where the at-risk revenue sits

On the forward book, the same concentration appears across segments. The comparison below sets each segment's current at-risk MRR against the monthly value it has already lost to churn. SMB carries the largest current at-risk MRR at $40,470, reflecting its size, while Enterprise, despite churning the fewest accounts, still carries $13,898 of at-risk MRR because each Enterprise account is worth so much that even a handful in distress represents real money. This is why value has to sit alongside risk in the prioritisation: a single wavering Enterprise account can outweigh a dozen Startup accounts.

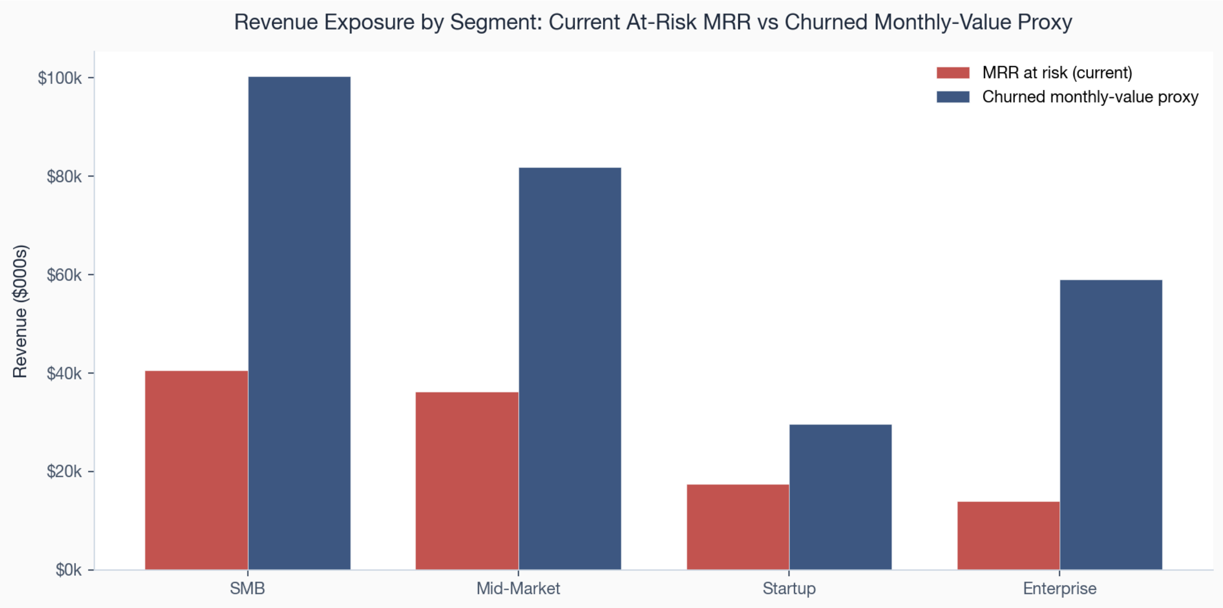

Figure 16. Current at-risk MRR against churned monthly value by segment. SMB leads on current exposure; Enterprise punches above its account count on value per account.

Segment	At-risk accts	At-risk MRR	Churned accts	Churned monthly value
Enterprise	17	$13,898	89	$59,019
Mid-Market	95	$36,189	227	$81,802
SMB	231	$40,470	456	$100,364
Startup	120	$17,374	211	$29,600

The risk tiers

The risk index sorts the 2,517 active accounts into four tiers. The critical tier holds 34 accounts billing $14,487 a month, the high tier 193 accounts billing $111,247, and the medium tier 487 accounts billing $149,818. The critical and high tiers together, 227 accounts and $125,734 of MRR, are small enough for a team to work by hand and large enough to matter. They are the spine of the priority queue.

These two upper tiers represent only 9.0 percent of scored accounts but 11.2 percent of scored-book MRR. That is not a perfect concentration ratio, but it is enough to justify named-account management. The more important point is capacity: two hundred twenty-three accounts can be assigned, worked, and tracked without turning the programme into a mass campaign. The medium tier should be monitored and largely automated; the low tier should be protected from unnecessary outreach so customer-facing teams do not create work where the data shows no current problem.

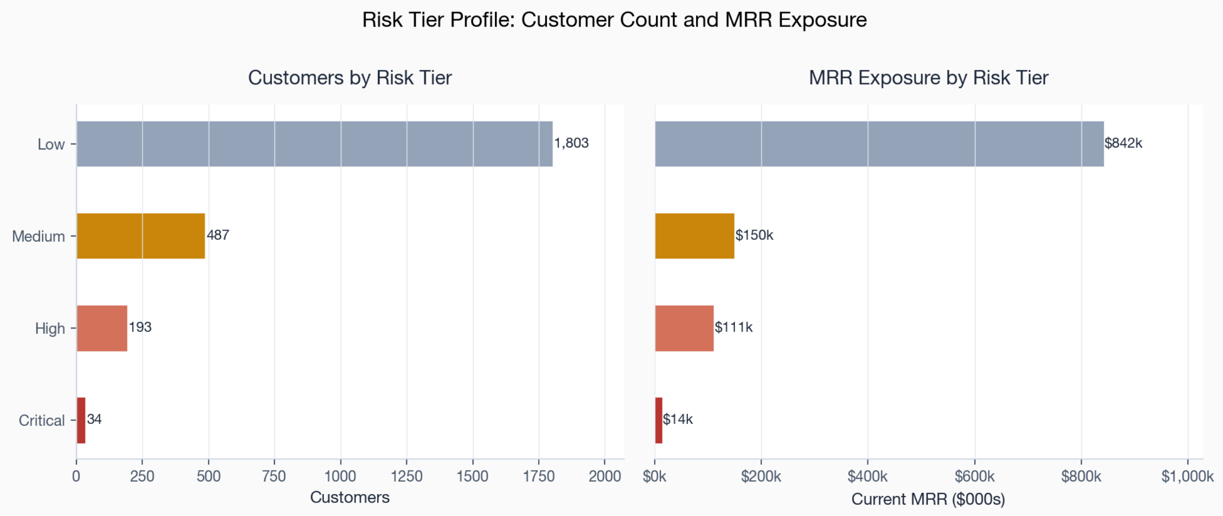

Figure 17. Account count and MRR exposure by risk tier. The critical and high tiers are a workable number of accounts holding a material share of exposed revenue.

The score distribution behind those tiers confirms that the book is low-risk in aggregate and the risk is a tail, not a bulk. The low tier dominates the count and clusters at the bottom of the score range; the critical and high tiers are a thin, well-separated tail at the top. That shape is what a prioritisation index should produce: it avoids labelling half the book as at risk and isolates the minority that needs review.

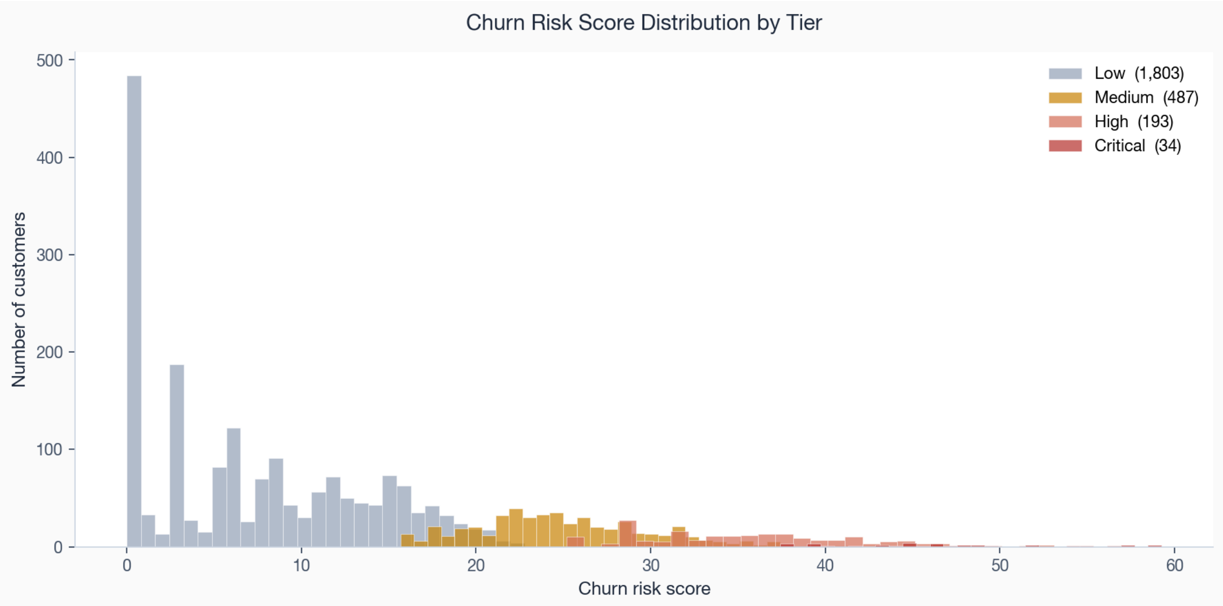

Figure 18. Distribution of churn risk scores by tier. The mass sits in the low-risk tier; the at-risk tiers form a distinct upper tail.

SECTION 10

Risks, limitations, and caveats

A report that hides its weaknesses is less useful than one that names them, because the reader cannot judge how far to trust a finding without knowing what could undermine it. Six limitations bound everything above and each changes how a specific finding should be read.

The data is synthetic

The book is generated from a fixed seed, not collected from a live business. Its relationships are designed and deterministic, so the magnitudes are illustrative rather than externally observed. The behavioural lifts in Section 08 are larger than a real book would show, because the comparison group of low-signal accounts almost never churns here. The method, the pipeline, the metric definitions, and the prioritisation logic are reusable starting points for real data, but source mappings, thresholds, validation rules, and outcome calibration would need to be fitted to the operating environment.

Recent acceleration is not yet a trend

The last three complete months have a higher realised churn rate than the prior nine-month baseline, with the final month at 1.4 percent. Three observations are not enough to separate a persistent shift from ordinary variation or a change in customer mix. The appropriate response is to monitor the next complete periods and decompose the change by segment, channel, plan, and tenure rather than declare a churn crisis.

Economics use recurring-value and direct-cost proxies

The MRR bridge reconciles contract-value movements, direct service costs, acquisition spend, payback, and capped 24-month LTV. These measures are applied consistently and support retention-adjusted comparisons, but they are not GAAP revenue recognition, audited cost allocation, remaining contract value, or a perpetuity. Absolute exposure remains a monthly run-rate proxy.

Relationships are associative

No finding in this report establishes cause. A low net promoter score does not necessarily cause a departure; it co-occurs with it. The intervention queues are therefore assigned with an explicit randomized holdout. The bundled forward outcomes are simulated to demonstrate the estimator and must not be presented as observed treatment efficacy. Weighted exposure sizes the scope; only live controlled outcomes can establish incremental value.

Tiering depends on the chosen thresholds

The value tiers and risk cut-points are quantile-based and would need recalibration on a real distribution. Moving a threshold moves accounts between tiers and changes the headline counts. The ranking is stable to these choices; the exact tier populations are not, and any operational commitment to a tier size should be set against real data.

The operating history is simulated

The system now retains monthly customer probabilities, tier transitions, calibration outcomes, and drift. That resolves the single-snapshot engineering gap, but the history is still generated rather than observed in operations. A live deployment needs enough complete monthly intervals to distinguish seasonality, recovery, definition changes, and persistent deterioration before alert thresholds or retraining cadence are treated as stable.

What would change the conclusion

The limitations above do not invalidate the recommendations, but they define the tests that would change them. The table below is the report's decision control: if one of these checks fails on real or refreshed data, the operating response should change before more budget is committed.

Conclusion at risk	Check to run	Decision trigger
Low-end acquisition is the main churn issue	Hold segment, plan, and region constant and re-estimate channel retention on live cohorts.	If Affiliate and Paid Search no longer over-index after controls, shift from budget reallocation to onboarding/product fixes.
Recent cohorts are deteriorating	Refresh the cohort read after the next three complete months and compare at fixed ages.	If the fixed-age gap closes, treat the spike as right-censoring noise rather than a structural mix shift.
Behavioural signals are intervention-ready	Run held-out tests for payment rescue, adoption reactivation, and service recovery.	If treated accounts do not outperform controls, downgrade the driver from an intervention lever to a monitoring signal.
The save desk should be value-weighted	Compare saved MRR, saved logos, and cost-to-serve by queue.	If small-account saves have materially better economics, rebalance capacity toward scaled automation rather than named-account coverage.

SECTION 11

Recommendations and action priorities

The findings converge on a short, ordered list. The ordering is by weighted exposure and operational separability, and each recommendation names the finding it rests on, the population it targets, and the metric that would prove it worked.

The four intervention queues that anchor the priorities are sized below. Read the scope column as current MRR passing through each queue. Weighted exposure multiplies that MRR by the target group's historical churn share, which puts the queues on a common ranking scale without estimating saved revenue. The gap between the two columns reflects the historical churn rate of the targeted accounts: the renewal save desk works a large book at a moderate rate, while payment rescue works a small book at a very high rate. The data ranks exposure; it does not contain implementation cost and therefore does not estimate return on effort.

Intervention queue	Accounts	MRR in scope	Hist. churn	Weighted exposure
Renewal Save Desk	451	$170,661	36%	$61,278
Service Recovery	549	$137,902	39%	$54,407
Payment Rescue	111	$27,970	72%	$20,247
Adoption Reactivation	111	$15,356	44%	$6,699

P1 Stand up a renewal save desk for Mid-Market and SMB

This play carries the largest weighted exposure in the book at $61,278, across 451 accounts holding $170,661 of MRR in scope, with the focus on Mid-Market and SMB where the value and the volume meet (Section 09, Figure 1). Work the critical and high risk tiers first, lead with the accounts flagged on sentiment and support, and run a pre-renewal save playbook with tailored offers. Measure success as the saved MRR among contacted accounts against a held-out control, not as gross contact volume.

P1 Fix the payment-failure path

Accounts with a failed payment churn at 72 percent, more than three times the rate of accounts that pay cleanly (Section 08). The payment-rescue queue is only 549 accounts but carries $54,407 of weighted exposure. The operating fix is straightforward: dunning sequencing, card-update prompts, and retry logic. It is the smallest of the high-exposure specialist queues and can run in parallel with the save desk. Measure recovered MRR per failed-payment account.

P2 Rebalance acquisition spend away from the weakest channels

Affiliate and Paid Search accounts churn at 41.6 and 34.3 percent against 18.3 percent for Partner and 28.0 percent for Referral (Section 07, Figure 10). Run a controlled marginal reallocation and retain it only if six-month retention improves after accounting for CAC, segment, plan, and acquisition volume.

P2 Run an adoption-reactivation programme for the usage-decline tail

Usage decline touches 1,580 accounts, the largest behavioural population, at a lower per-account certainty than the sentiment flags (Section 08). It suits a scaled, automated motion: onboarding refreshers, feature nudges, and success check-ins. The reactivation queue carries $20,247 of weighted exposure across 111 accounts. Measure the share of flagged accounts that return to a positive usage trend.

P3 Treat low-tier and Basic-plan signups as a distinct, instrumented class

Basic-plan accounts churn at 38.1 percent (Section 07). Rather than a save motion, this calls for a product and onboarding fix: a guided first-value path for entry-tier accounts and clearer upgrade prompts for the ones that show real engagement. Measure the rate at which Basic accounts either reach an activation milestone or upgrade within ninety days.

P3 Replace simulated intervention outcomes with observed results

The controlled design already assigns 172 accounts to treatment and 171 to holdout with maximum baseline SMD 0.105. Preserve that assignment, record delivery and cost, and replace simulated outcomes with observed 90-day churn before scaling or claiming ROI.

Sequenced this way, the first two actions address open-account exposure, the middle two test acquisition and adoption hypotheses, and the last two build measurement discipline. Capacity, implementation cost, and delivery timing are not present in the dataset and must be set by the operating team.

Sequencing, ownership, and success measures

The plan resolves to the following sequence. Each action carries one owner, one horizon, and one measure that settles whether it worked, so the programme can be governed on outcomes rather than activity.

#	Action	Owner	Horizon	Success measure
P1	Renewal save desk	Customer Success	0-90 days	Saved MRR vs held-out control
P1	Payment-failure path	Billing / RevOps	0-60 days	Recovered MRR per failed-payment account
P2	Acquisition rebalance	Marketing	1-2 quarters	6-month retention of new cohorts by channel
P2	Adoption reactivation	Customer Success	1-2 quarters	Share of flagged accounts returning to positive usage
P3	Basic-plan onboarding fix	Product	2 quarters	90-day activation or upgrade rate
P3	Driver experiments	Analytics	Ongoing	Measured lift vs control per queue

Management cadence and escalation rules

The programme should be governed as a short operating cycle, not a one-off analysis handoff. The first review establishes the queue, owners, and controls. The second tests the mechanics. The third decides which motions scale and which are stopped. This cadence keeps the team from mistaking activity for retention impact.

Review point	Decision to make	Evidence required
Day 30	Confirm that the critical/high-risk queue is assigned and contactable.	Named-account coverage, payment-failure retry status, and baseline control groups for each intervention.
Day 60	Decide whether payment rescue and adoption reactivation are clearing their minimum return bar.	Recovered MRR per failed-payment account, usage trend recovery rate, and untreated-control comparison.
Day 90	Scale, stop, or redesign the save-desk motion.	Saved MRR versus control, renewal outcomes by risk tier, and cost per account worked.
Quarterly	Reallocate acquisition budget by observed cohort quality.	Six-month retention by channel and plan, plus finance-approved CAC, gross margin, and payback.

The four intervention queues currently cover $351,889 of MRR in scope and $142,631 of weighted monthly exposure. That is the starting control total. A good first quarter does not need to eliminate the risk; it needs to show that the critical and high-risk tail is shrinking, that payment failures are being recovered faster, and that new cohorts are no longer over-weighted toward the weakest channels. Those are the proof points that the recommendation is working.

SECTION 12

Production evidence still required

The engineering gaps identified in the initial analysis are now implemented: reconciled MRR movements, unit economics, calibrated temporal probabilities, randomized holdouts, incremental saved-MRR estimation, and tier-transition alerts. The remaining questions are evidence gaps that only live operating history can close.

Implemented control	Current release evidence	Production evidence needed
Retention-adjusted unit economics	Average NRR 99.3%; direct-cost margin 74.3%; blended CAC $951.	Finance-approved spend allocation, cost-to-serve, revenue recognition, and mature cohort follow-up.
Point-in-time probability model	Out-of-time ROC AUC 0.823, average precision 0.312, Brier 0.0344.	Live outcomes, stable feature definitions, independent recalibration, and customer-cluster uncertainty estimates.
Product activation definition	Trailing sessions and adoption separate risk, but do not identify a causal first-value event.	Event-level product usage mapped to activation, expansion, support volume, and churn outcomes.
Randomized intervention holdout	172 treatment and 171 control; maximum baseline SMD 0.105.	Observed delivery, treatment cost, 90-day outcomes, and sufficient sample size before efficacy or ROI claims.
Risk migration and alerts	Monthly tier transitions, outcome calibration, PSI, and 0 open release alerts.	Enough complete operating months to estimate seasonality, alert precision, recovery, and retraining cadence.

The simulated intervention produces an incremental saved-MRR estimate of -$131, with a 95 percent interval from -$10,907 to $10,645. That number validates the holdout estimator and reconciliation logic; budget allocation must wait for observed outcomes and delivery cost.

SECTION 13

Appendix

A. Cumulative churn share by commercial cut

Cut	Group	Accounts	Churned	Churn share	Avg MRR
Segment	Enterprise	536	89	16.6%	$910
Segment	Mid-Market	974	227	23.3%	$466
Segment	SMB	1,416	456	32.2%	$246
Segment	Startup	574	211	36.8%	$170
Plan	Enterprise	371	48	12.9%	$1,659
Plan	Pro	966	207	21.4%	$514
Plan	Growth	1,207	364	30.2%	$180
Plan	Basic	956	364	38.1%	$62
Channel	Affiliate	392	163	41.6%	$249
Channel	Paid Search	823	282	34.3%	$315
Channel	Referral	378	106	28.0%	$401
Channel	Organic	777	190	24.5%	$388
Channel	Outbound	634	151	23.8%	$492
Channel	Partner	496	91	18.3%	$536
Region	LATAM	555	175	31.5%	$384
Region	APAC	488	141	28.9%	$366
Region	Europe	1,090	298	27.3%	$411
Region	North America	1,367	369	27.0%	$401

B. Full behavioural driver detail

Signal	Accounts	Churn w/ signal	Churn w/o signal	Lift
Failed Payment	402	72%	22%	3.2×
Low NPS	879	47%	22%	2.2×
High Support Ticket	1,211	40%	22%	1.8×
Low Feature Adoption	875	40%	24%	1.7×
Usage Decline	1,580	35%	23%	1.5×

C. Recommended-action distribution across the scored book

Recommended action	Accounts	Share of scored book
monitor only	1,884	74.9%
renewal conversation	457	18.2%
product adoption campaign	83	3.3%
customer success outreach	79	3.1%
executive save motion	9	0.4%
billing intervention	5	0.2%

D. Monthly retention trend, last twelve months

The active book at the start of each month, the accounts and revenue lost during it, and the resulting churn rates. The customer and revenue churn columns are the data behind Figures 2 and 3; the right-censoring caveat in Section 10 applies to the final rows.

Month	Active accts	Active MRR	Churned	Cust. churn	Rev. churn
Mar 2025	2,046	$873k	31	1.5%	0.8%
Apr 2025	2,092	$896k	33	1.6%	0.8%
May 2025	2,137	$912k	33	1.5%	1.4%
Jun 2025	2,177	$938k	31	1.4%	1.0%
Jul 2025	2,209	$956k	39	1.8%	0.9%
Aug 2025	2,257	$986k	30	1.3%	0.8%
Sep 2025	2,292	$1,006k	20	0.9%	0.3%
Oct 2025	2,333	$1,028k	35	1.5%	0.8%
Nov 2025	2,363	$1,047k	34	1.4%	1.2%
Dec 2025	2,405	$1,067k	46	1.9%	0.9%
Jan 2026	2,433	$1,075k	37	1.5%	0.8%
Feb 2026	2,467	$1,085k	35	1.4%	0.8%

E. Figure index

Figures 1 through 18 are the static export of the chart pack in `outputs/graphs/`, regenerated deterministically from the same processed data and analytical tables that drive the live dashboard. The dashboard (`executive-retention-command-center.html`) carries the interactive versions of the trend, segment, driver, and queue views, with period, segment, and channel filters. This report and that dashboard are two renderings of one governed pipeline; neither contains a number the other cannot reproduce.

F. Per-account feature dictionary

The behavioural signals examined in Sections 07 and 08 and scored in the risk index are measured at the account's churn date or, for active accounts, at the snapshot. The principal fields are defined below.

Field	Definition
`tenure_days`	Days from subscription start to churn or snapshot.
`current_mrr`	Account's monthly recurring revenue at observation.
`usage_trend`	Mean weekly sessions in days 0-29 minus the mean in days 30-59.
`recent_sessions_30d / 90d`	Product sessions in the trailing window.
`feature_adoption_score_recent`	Mean feature-adoption score in the trailing 30 days, 0 to 100.
`support_tickets_30d / 90d`	Support tickets filed in the window.
`nps_score_recent`	Mean NPS in the trailing 90 days.
`failed_payments_90d`	Count of failed billing attempts in 90 days.
`payment_failure_flag`	Any failed payment in the window.
`renewal_near_flag`	Contract renewal falls within 45 days.
`at_risk_flag`	Source subscription status is at_risk.
`churn_flag`	Account has closed (the analysis target).

G. Strategic expansion release evidence

Control	Release result	Interpretation
Retention economics	NRR 99.3%; gross-margin proxy 74.3%; CAC $951	Complete periods only; recurring-value and direct-cost proxies.
90-day probability model	ROC AUC 0.823; AP 0.312; Brier 0.0344	Independent calibration and out-of-time evaluation on synthetic data.
Intervention holdout	172 treatment / 171 control; max SMD 0.105	Assignment is valid; forward outcomes are a simulation.
Incremental saved MRR	-$131; 95% interval -$10,907 to $10,645	Estimator demonstration, not production efficacy or ROI.
Monitoring	0 open alerts; 57 / 57 validation checks pass	Monthly transitions exclude partial periods from alert decisions.